

Thermo Scientific™ HyPerforma™ S.U.F. 300 L - 60-300 L working volume

thermo-suf-300

Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	60 L	Source-supported
Maximum working volume	300 L	Source-supported
Nominal vessel / bag size	300 L	Source-supported
Total vessel volume loaded into the form	435 L	Source-supported
Initial working volume loaded into the form	60 L	Derived / estimated
Product-contact material	Plastic / single-use	App assumption
Height : diameter	2	Source-supported
Internal / chamber diameter	0.572 m	Source-supported
Liquid height at maximum working volume	1.23 m	Source-supported
Impeller / mixing type	Six-blade Rushton turbine	Source-supported
Impeller count	3	Source-supported
Impeller diameter	0.1879 m	Source-supported
Power number	4.1	Source-supported
Initial agitation	100 rpm	App assumption
Maximum agitation	120 rpm	App assumption
Initial aeration	0.5 vvm	App assumption
Maximum aeration	2 vvm	App assumption
Baffled vessel	Yes	Source-supported
Maximum pressure	0.03 bar(g)	Source-supported
Default operating pressure	0 bar(g)	App assumption
Maximum inlet oxygen fraction	100%	App assumption

Parameter evidence and derivations

Minimum working volume - 60 L

Source-supported | Thermo HyPerforma single-use fermentor user guide (publisher Dutch URL) - PDF page 116; Table 4.1, printed page 111

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 60 L.

Limitations: Published vessel diameter (not separate BPC chamber diameter), liquid height, three 6-blade Rushton turbines and four baffles. Power number is an average over 20-100% RPM.

Local original: [Thermo HyPerforma single-use fermentor user guide \(publisher Dutch URL\) - PDF page 116](#)

[Manufacturer publication](#)

Maximum working volume - 300 L

Source-supported | Thermo HyPerforma single-use fermentor user guide (publisher Dutch URL) - PDF page 116; Table 4.1, printed page 111

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 300 L.

Limitations: Published vessel diameter (not separate BPC chamber diameter), liquid height, three 6-blade Rushton turbines and four baffles. Power number is an average over 20-100% RPM.

Local original: [Thermo HyPerforma single-use fermentor user guide \(publisher Dutch URL\) - PDF page 116](#)

[Manufacturer publication](#)

Nominal vessel / bag size - 300 L

Source-supported | Thermo HyPerforma single-use fermentor user guide (publisher Dutch URL) - PDF page 116; Table 4.1, printed page 111

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 300 L.

Limitations: Published vessel diameter (not separate BPC chamber diameter), liquid height, three 6-blade Rushton turbines and four baffles. Power number is an average over 20-100% RPM.

Local original: [Thermo HyPerforma single-use fermentor user guide \(publisher Dutch URL\) - PDF page 116](#)

[Manufacturer publication](#)

Total vessel volume loaded into the form - 435 L

Source-supported | Thermo HyPerforma single-use fermentor user guide (publisher Dutch URL) - PDF page 116; Table 4.1, printed page 111

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 435 L.

Limitations: Published vessel diameter (not separate BPC chamber diameter), liquid height, three 6-blade Rushton turbines and four baffles. Power number is an average over 20-100% RPM.

Local original: [Thermo HyPerforma single-use fermentor user guide \(publisher Dutch URL\) - PDF page 116](#)

[Manufacturer publication](#)

Initial working volume loaded into the form - 60 L

Derived / estimated | Form-population rule

Rule: initial working volume = min(minimum working volume, maximum working volume)

Selection / calculation: min(60 L, 300 L) = 60 L

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - Plastic / single-use

App assumption | Unverified legacy preset

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: Plastic / single-use.

Limitations: No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Height : diameter - 2

Source-supported | Thermo HyPerforma single-use fermentor user guide (publisher Dutch URL) - PDF page 116; Table 4.1, printed page 111

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 2.

Limitations: Published vessel diameter (not separate BPC chamber diameter), liquid height, three 6-blade Rushton turbines and four baffles. Power number is an average over 20-100% RPM.

Local original: [Thermo HyPerforma single-use fermentor user guide \(publisher Dutch URL\) - PDF page 116](#)

Manufacturer publication

Internal / chamber diameter - 0.572 m

Source-supported | Thermo HyPerforma single-use fermentor user guide (publisher Dutch URL) - PDF page 116; Table 4.1, printed page 111

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.572 m.

Limitations: Published vessel diameter (not separate BPC chamber diameter), liquid height, three 6-blade Rushton turbines and four baffles. Power number is an average over 20-100% RPM.

Local original: [Thermo HyPerforma single-use fermentor user guide \(publisher Dutch URL\) - PDF page 116](#)

Manufacturer publication

Liquid height at maximum working volume - 1.23 m

Source-supported | Thermo HyPerforma single-use fermentor user guide (publisher Dutch URL) - PDF page 116; Table 4.1, printed page 111

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 1.23 m.

Limitations: Published vessel diameter (not separate BPC chamber diameter), liquid height, three 6-blade Rushton turbines and four baffles. Power number is an average over 20-100% RPM.

Local original: [Thermo HyPerforma single-use fermentor user guide \(publisher Dutch URL\) - PDF page 116](#)

Manufacturer publication

Impeller / mixing type - Six-blade Rushton turbine

Source-supported | Thermo HyPerforma single-use fermentor user guide (publisher Dutch URL) - PDF page 116; Table 4.1, printed page 111

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Six-blade Rushton turbine.

Limitations: Published vessel diameter (not separate BPC chamber diameter), liquid height, three 6-blade Rushton turbines and four baffles. Power number is an average over 20-100% RPM.

Local original: [Thermo HyPerforma single-use fermentor user guide \(publisher Dutch URL\) - PDF page 116](#)

Manufacturer publication

Impeller count - 3

Source-supported | Thermo HyPerforma single-use fermentor user guide (publisher Dutch URL) - PDF page 116; Table 4.1, printed page 111

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 3.

Limitations: Published vessel diameter (not separate BPC chamber diameter), liquid height, three 6-blade Rushton turbines and four baffles. Power number is an average over 20-100% RPM.

Local original: [Thermo HyPerforma single-use fermentor user guide \(publisher Dutch URL\) - PDF page 116](#)

Manufacturer publication

Impeller diameter - 0.1879 m

Source-supported | Thermo HyPerforma single-use fermentor user guide (publisher Dutch URL) - PDF page 116; Table 4.1, printed page 111

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.1879 m.

Limitations: Published vessel diameter (not separate BPC chamber diameter), liquid height, three 6-blade Rushton turbines and four baffles. Power number is an average over 20-100% RPM.

Local original: [Thermo HyPerforma single-use fermentor user guide \(publisher Dutch URL\) - PDF page 116](#)

[Manufacturer publication](#)

Power number - 4.1

Source-supported | Thermo HyPerforma single-use fermentor user guide (publisher Dutch URL) - PDF page 116; Table 4.1, printed page 111

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 4.1.

Limitations: Published vessel diameter (not separate BPC chamber diameter), liquid height, three 6-blade Rushton turbines and four baffles. Power number is an average over 20-100% RPM.

Local original: [Thermo HyPerforma single-use fermentor user guide \(publisher Dutch URL\) - PDF page 116](#)

[Manufacturer publication](#)

Initial agitation - 100 rpm

App assumption | Vessel-specific preset override

Rule: $\text{baseRpm} = 100$

Selection / calculation: No calculation. The value is written directly into this vessel's catalogue entry.

Limitations: This vessel-specific value overrides the generic microbial stirred-tank profile. It is still treated as an app assumption unless the parameter is explicitly listed as source-supported. This is the initial agitation setting loaded into the form.

Maximum agitation - 120 rpm

App assumption | Thermo HyPerforma single-use fermentor user guide (publisher Dutch URL) - PDF page 102; Maximum speed by working volume

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 120 rpm.

Limitations: This simulator has one constant speed ceiling. Conservatively selected the published limit at minimum working volume; it does not implement the full fill-dependent envelope. At full volume the rated ceiling is 375 rpm.

Local original: [Thermo HyPerforma single-use fermentor user guide \(publisher Dutch URL\) - PDF page 102](#)

[Manufacturer publication](#)

Initial aeration - 0.5 vvm

App assumption | Vessel-specific preset override

Rule: $\text{baseVvm} = 0.5$

Selection / calculation: No calculation. The value is written directly into this vessel's catalogue entry.

Limitations: This vessel-specific value overrides the generic microbial stirred-tank profile. It is still treated as an app assumption unless the parameter is explicitly listed as source-supported. This is the initial gas rate loaded into the form.

Maximum aeration - 2 vvm

App assumption | Vessel-specific preset override

Rule: $\text{maxVvm} = 2$

Selection / calculation: No calculation. The value is written directly into this vessel's catalogue entry.

Limitations: This vessel-specific value overrides the generic microbial stirred-tank profile. It is still treated as an app assumption unless the parameter is explicitly listed as source-supported. This is the gas-rate ceiling available to the generic DO-control cascade.

Baffled vessel - Yes

Source-supported | Thermo HyPerforma single-use fermentor user guide (publisher Dutch URL) - PDF page 116; Table 4.1, printed page 111

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Yes.

Limitations: Published vessel diameter (not separate BPC chamber diameter), liquid height, three 6-blade Rushton turbines and four baffles. Power number is an average over 20-100% RPM.

Local original: [Thermo HyPerforma single-use fermentor user guide \(publisher Dutch URL\) - PDF page 116](#)

Manufacturer publication

Maximum pressure - 0.03 bar(g)

Source-supported | Thermo HyPerforma single-use fermentor user guide (publisher Dutch URL) - PDF page 21; Section 1.4.2 Operating pressure, printed p. 16

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.03 bar(g).

Limitations: Manufacturer prints 0.03 bar (0.5 psi). Use the explicit bar limit; the parenthetical conversion is rounded. BPC operating limit, not jacket pressure or vessel design rating.

Local original: [Thermo HyPerforma single-use fermentor user guide \(publisher Dutch URL\) - PDF page 21](#)

Manufacturer publication

Default operating pressure - 0 bar(g)

App assumption | Thermo HyPerforma single-use fermentor user guide (publisher Dutch URL) - PDF page 21; Section 1.4.2

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 0 bar(g).

Limitations: Atmospheric app starting value replaces 0.1 bar, which exceeds the published BPC limit. Not a manufacturer process setpoint.

Local original: [Thermo HyPerforma single-use fermentor user guide \(publisher Dutch URL\) - PDF page 21](#)

Manufacturer publication

Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: maxOxygenFraction = 100

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

The 3:1 reactor H/D, three Rushton-style impellers and 375 rpm maximum are model-specific source-supported fields.

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

Speed envelope, PDF p. 102: 60 L: 120 rpm; 150 L: 240 rpm; 300 L: 375 rpm. Volume-dependent agitation requires a future model extension; the conservative preset is intentionally restrictive.

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Local original document identities

Thermo HyPerforma single-use fermentor user guide (publisher Dutch URL)

thermo-hyperforma-single-use-fermentor-user-guide-publisher-dutch-url-32e1e2f2165b.pdf

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